

[University Of Petroleum and Energy Studies]

Department of Computer Science and Engineering

LUNG CANCER MALIGNANCY PREDICTION SYSTEM

Dual Pipeline: Tabular + Image Deep Learning

Comprehensive Project Report

Bachelor of Technology
in
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Dataset: DICOM-LIDC-IDRI-Nodules (The Cancer Imaging Archive)

Live Application: <https://lung--cancer--prediction.streamlit.app>

Github Repository Link:

https://github.com/vaidaadya19-prog/Lung_Cancer_Prediction

Abstract

Lung cancer remains one of the leading causes of cancer-related mortality worldwide, with poor prognosis largely attributable to late-stage diagnosis. Early and accurate classification of pulmonary nodules as malignant or benign is critical for improving patient survival outcomes. This project presents a comprehensive dual-pipeline artificial intelligence system for lung nodule malignancy prediction, developed using the Lung Imaging Database Consortium and Image Database Resource Initiative (LIDC-IDRI) dataset — the most comprehensive publicly available annotated lung CT imaging repository.

The system integrates two complementary and independently evaluated pipelines. The first is a tabular machine learning pipeline that leverages radiomic features extracted from radiologist annotations encoded in DICOM Structured Report (SR) files. The second is a deep learning image pipeline employing a ResNet18 convolutional neural network trained on nodule segmentation masks derived from DICOM Segmentation (SEG) objects. Three class-imbalance mitigation strategies — Synthetic Minority Oversampling Technique (SMOTE), BorderlineSMOTE, and Conditional Tabular Generative Adversarial Network (CTGAN) — are implemented and rigorously compared.

Explainability is a central design objective: SHAP (SHapley Additive exPlanations) TreeExplainer provides interpretable feature importance for the tabular pipeline, while Gradient-weighted Class Activation Mapping (Grad-CAM) offers spatial attribution visualisations for the image pipeline. The tabular pipeline achieves an AUC-ROC of 0.880 and a recall of 0.859, significantly outperforming the image pipeline (AUC 0.740), primarily because the segmentation masks encode shape rather than texture-rich CT intensity information.

The complete system is deployed as a publicly accessible live web application built with Streamlit, enabling clinicians and researchers to submit either radiomic feature scores or nodule images and receive real-time malignancy predictions accompanied by confidence scores and visual explanations. The project demonstrates the feasibility of AI-assisted radiological decision support and establishes a reproducible foundation for future research.

Keywords: Lung cancer, LIDC-IDRI, malignancy prediction, radiomic features, ResNet18, SMOTE, CTGAN, SHAP, Grad-CAM, Streamlit, deep learning, medical imaging.

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Chapter 1

Introduction

1.1 Background

Lung cancer is the most lethal malignancy globally, accounting for approximately 18% of all cancer-related deaths. The five-year survival rate for lung cancer patients remains as low as 19%, a figure largely driven by the frequency of late-stage diagnoses. When detected at Stage I, however, survival rates rise to over 70%, underscoring the life-saving potential of early detection.

Computed Tomography (CT) scanning is the primary diagnostic modality for identifying pulmonary nodules. Manual radiological review of CT scans is time-consuming, subjective, and susceptible to inter-radiologist variability. The Lung Imaging Database Consortium and Image Database Resource Initiative (LIDC-IDRI) dataset addresses this challenge by providing a large-scale, expert-annotated collection of lung CT scans with consensus ratings from four independent thoracic radiologists per nodule. This dataset has become the de facto standard for lung nodule research and provides the data foundation for the present project.

1.2 Motivation

The clinical need for automated, reliable, and interpretable tools for lung nodule characterisation is well established. While numerous research systems have been proposed in the literature, many suffer from limited explainability, inadequate handling of class imbalance, or lack of deployment as accessible applications. This project is motivated by the goal of building an end-to-end system that not only achieves strong predictive performance but also provides transparent, interpretable outputs suitable for supporting radiologists in clinical decision-making.

1.3 Problem Statement

The central challenge addressed by this project is to design, implement, and deploy an accurate, explainable, and publicly accessible system that classifies lung nodules as malignant or benign using either radiologist-assessed radiomic features or nodule segmentation images. The system must additionally address the class imbalance inherent in medical datasets

— where benign nodules significantly outnumber malignant ones — and provide visual explanations to promote trust and interpretability.

1.4 Objectives

The specific objectives of this project are as follows:

- Develop a tabular machine learning pipeline using radiomic features extracted from DICOM SR files, with multiple imbalance mitigation strategies (SMOTE, BorderlineSMOTE, CTGAN).
- Develop a deep learning image pipeline employing a ResNet18 CNN trained on nodule segmentation patches extracted from DICOM SEG files.
- Implement and compare three imbalance handling strategies in terms of their effect on AUC-ROC, Recall, F1-Score, and Precision.
- Provide model explainability through SHAP feature importance analysis (tabular pipeline) and Grad-CAM spatial attribution visualisations (image pipeline).
- Deploy the complete system as a live, publicly accessible web application using Streamlit.
- Conduct a rigorous comparative analysis of both pipelines to identify strengths, limitations, and directions for future improvement.

Chapter 2

Literature Review

The automated analysis of pulmonary nodules has attracted substantial research attention over the past two decades, driven by advances in machine learning and the availability of annotated imaging datasets.

2.1 Lung Nodule Datasets and Annotation

Armato et al. (2011) introduced the LIDC-IDRI dataset, which remains the most widely used benchmark for lung nodule research. The dataset provides expert annotations from four radiologists per nodule, including nine radiomic characteristics and a malignancy suspicion score on a five-point scale. Fedorov et al. (2020) subsequently published the DICOM re-encoding of these annotations — the DICOM-LIDC-IDRI-Nodules version used in the present project — converting XML-formatted radiologist annotations into standardised DICOM SR and SEG objects, facilitating programmatic access without requiring the full 133 GB CT scan archive.

2.2 Machine Learning Approaches for Nodule Classification

Hancock and Magnan (2016) demonstrated that ensemble-based classifiers trained on quantified radiomic features extracted directly from radiologist scores — without raw image processing — could achieve clinically useful performance. Their work established that features such as spiculation, lobulation, and texture carry strong discriminative value for malignancy prediction, a finding corroborated in the present study’s SHAP analysis.

Classical approaches including Support Vector Machines, Random Forests, and Gradient Boosting Machines have been widely applied to tabular radiomic features. Gradient Boosting Machines have demonstrated consistently strong AUC performance on structured medical data due to its ability to capture non-linear interactions among features.

2.3 Deep Learning and Convolutional Neural Networks

He et al. (2016) introduced Deep Residual Networks (ResNets), demonstrating that residual connections enable the training of very deep networks without degradation. ResNet18 and ResNet50, pretrained on ImageNet, have been widely adopted for medical image classification via transfer learning. This approach is particularly advantageous when labelled medical imaging data is scarce, as pretrained weights provide useful low-level feature detectors that can be fine-tuned on domain-specific data.

2.4 Class Imbalance Handling

Class imbalance is a pervasive challenge in medical classification tasks. Chawla et al. (2002) introduced SMOTE, which generates synthetic minority class samples by interpolating between existing samples in feature space. Han et al. (2005) proposed BorderlineSMOTE, a refinement that focuses synthetic generation on borderline instances near the decision boundary, typically improving recall at the cost of precision.

Xu et al. (2019) proposed the Conditional Tabular GAN (CTGAN), a generative adversarial network architecture specifically designed for tabular data. CTGAN employs mode-specific normalisation and conditional vectors to handle the mixed discrete-continuous nature of tabular features, producing higher-fidelity synthetic samples than traditional oversampling approaches.

2.5 Explainable AI in Medical Imaging

Lundberg and Lee (2017) proposed SHAP, a unified framework for model interpretability based on Shapley values from cooperative game theory. TreeExplainer, a computationally efficient SHAP variant for tree-based models, provides exact Shapley value calculations and has been widely adopted for tabular medical data.

Selvaraju et al. (2017) introduced Grad-CAM, a gradient-based class activation mapping technique that generates spatial heatmaps indicating which regions of an input image most strongly influence a CNN's prediction. Grad-CAM has been extensively applied to medical imaging tasks including nodule analysis, pathology slide classification, and retinal disease detection.

Chapter 3

Dataset Description

3.1 LIDC-IDRI Overview

The Lung Imaging Database Consortium and Image Database Resource Initiative (LIDC-IDRI) is the largest publicly available annotated lung CT dataset. Collected under the sponsorship of the National Cancer Institute (NCI) through a collaboration of seven academic centres and eight medical imaging companies, the dataset provides a rigorous multi-radiologist annotation framework. Table 3.1 summarises the principal dataset properties.

Table 3.1: LIDC-IDRI Dataset Overview

Property	Details
Full Dataset Size	133.15 GB (DICOM CT scans)
Dataset Version Used	DICOM-LIDC-IDRI-Nodules (2.51 GB)
Total Patients	1,010 subjects
Patients with Nodule Annotations	875 subjects
Total Nodule Annotations	6,859 DICOM SR + 6,859 DICOM SEG objects
Distinct Nodules	2,651
Minimum Nodule Size	3 mm diameter
Radiologists per Nodule	4 independent expert thoracic radiologists
Source	The Cancer Imaging Archive (TCIA)
License	CC BY 3.0

3.2 Dataset Version Used

The DICOM-LIDC-IDRI-Nodules dataset (Version 2, updated September 2020) was used for this project. This standardised version converts the original XML-formatted radiologist annotations into DICOM Structured Report (SR) objects and DICOM Segmentation (SEG) objects, enabling programmatic parsing without requiring the complete 133 GB CT scan archive. The dataset was downloaded using the NBIA Data Retriever tool from The Cancer Imaging Archive.

3.3 Dataset Structure

After download and extraction, the dataset is organised hierarchically by patient, study date, and annotation object. Each patient folder contains paired SR and SEG files for each annotated nodule and radiologist annotation. The SR files encode the radiomic feature ratings, while the SEG files encode the corresponding three-dimensional binary segmentation masks.

3.4 File Types

Table 3.2 summarises the two DICOM file types used in this project and their respective roles in the dual-pipeline system.

Table 3.2: DICOM File Types and Their Purpose

File Type	Modality	Contents	Used For
DICOM SR	SR	Radiologist ratings of 9 nodule characteristics including malignancy score (1–5)	Tabular Pipeline
DICOM SEG	SEG	3D binary segmentation mask delineating the nodule volume	Image Pipeline

3.5 Radiomic Features

Each radiologist independently rates nine features for every annotated nodule on a numeric scale. Scores from all four radiologists are averaged to produce a consensus feature vector. Table 3.3 describes each feature and its scale.

Table 3.3: Radiomic Features Extracted from DICOM SR Files

Feature	Scale	Description
Subtlety	1–5	1 = Extremely subtle; 5 = Obvious nodule
Internal Structure	1–4	1 = Soft tissue; 2 = Fluid; 3 = Fat; 4 = Air
Calcification	1–6	1 = Absent; 6 = Eccentric calcification
Sphericity	1–5	1 = Linear shape; 5 = Round shape
Margin	1–5	1 = Poorly defined; 5 = Sharp margin
Lobulation	1–5	1 = No lobulation; 5 = Marked lobulation
Spiculation	1–5	1 = No spiculation; 5 = Marked spiculation
Texture	1–5	1 = Non-solid / Ground-Glass Opacity; 5 = Solid
Malignancy	1–5	1 = Highly unlikely malignant; 5 = Highly suspicious

3.6 Label Assignment

Binary class labels are derived from the consensus malignancy score. Nodules with a mean malignancy score of 4 or 5 are classified as malignant (label = 1), while scores of 1, 2, or 3 are classified as benign (label = 0). Table 3.4 presents the distribution of malignancy scores in the extracted dataset.

Table 3.4: Malignancy Score Distribution and Label Assignment

Malignancy Score	Count	Label	Interpretation
1	11	Benign (0)	Highly unlikely malignant
2	55	Benign (0)	Unlikely malignant
3	233	Benign (0)	Indeterminate
4	113	Malignant (1)	Moderately suspicious
5	36	Malignant (1)	Highly suspicious

3.7 Dataset Statistics After Processing

Table 3.5 summarises the dataset statistics following SR parsing and SEG patch extraction.

Table 3.5: Processed Dataset Statistics

Metric	Value
Total SR records extracted	448
Malignant nodules (label = 1)	149 (33.3%)
Benign nodules (label = 0)	299 (66.7%)
Unique patients	296
Total SEG patches extracted	3,219
Malignant patches	723 (22.5%)
Benign patches	2,496 (77.5%)
Patch size	64×64 pixels

Chapter 4

System Architecture

4.1 Overall Architecture

The system consists of two independent but complementary prediction pipelines, both sourced from the same LIDC-IDRI dataset. The tabular pipeline processes radiomic feature scores extracted from DICOM SR files, while the image pipeline processes nodule segmentation patches extracted from DICOM SEG files. Both pipelines are integrated into a single deployable web application. Table 4.1 provides a high-level comparison of the two pipelines.

Table 4.1: Dual-Pipeline Architecture Comparison

Component	Tabular Pipeline	Image Pipeline
Input	8 radiomic feature scores (1–5 scale)	Nodule segmentation patch (64×64 pixels)
Preprocessing	StandardScaler normalisation	Resize to 224×224; normalise to [0, 1]
Imbalance Handling	SMOTE / BorderlineSMOTE / CTGAN	Data augmentation (flip, rotate)
Model	Random Forest, Gradient Boosting, Logistic Regression	ResNet18 (pretrained on ImageNet)
Explainability	SHAP TreeExplainer	Grad-CAM
Best AUC	0.880	0.740
Best Recall	0.859	0.297

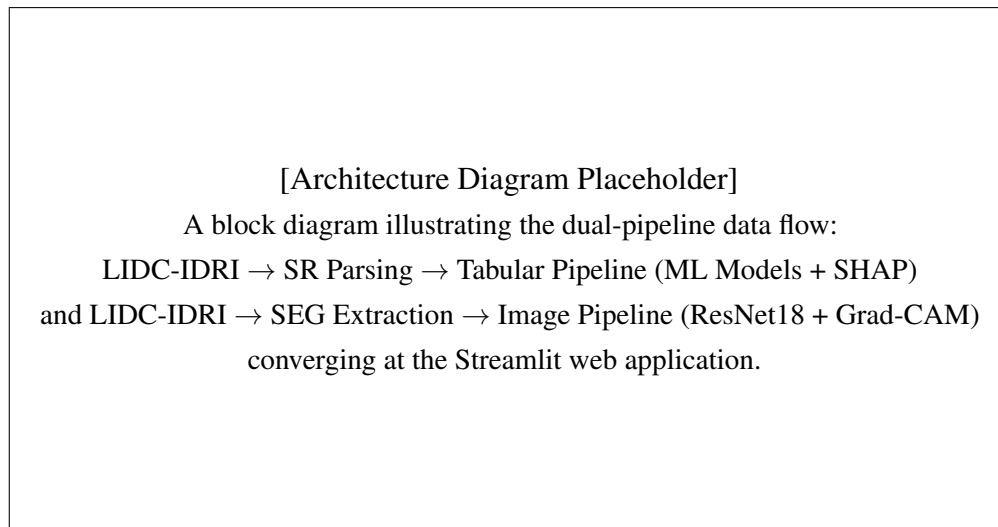


Figure 4.1: Overall System Architecture — Dual Pipeline

4.2 Technology Stack

Table 4.2 summarises the technologies and libraries used across all components of the system.

Table 4.2: Technology Stack

Category	Technology	Purpose
Language	Python 3.12	All processing and modelling
Environment	Google Colab (T4 GPU)	Training and development
Data Storage	Google Drive	Dataset and model persistence
DICOM Processing	pydicom	Reading .dcm files
ML Models	scikit-learn 1.4	Random Forest, GBM, Logistic Regression
Deep Learning	PyTorch 2.2 + torchvision	ResNet18 CNN training
Imbalance Handling	imbalanced-learn	SMOTE, BorderlineSMOTE
Synthetic Data	SDV (CTGANSynthesizer)	CTGAN oversampling
Explainability	SHAP	Feature importance
Image Processing	OpenCV	Patch extraction, resizing
Web Application	Streamlit	User interface
Model Hosting	GitHub + Google Drive	Model file storage
Deployment	Streamlit Cloud	Live web application

Chapter 5

Methodology and Implementation

5.1 Environment Setup

The project was developed in a Google Colab environment with a T4 GPU runtime. Key libraries including `pydicom`, `SimpleITK`, `scikit-image`, `imbalanced-learn`, `sdv`, and `shap` were installed programmatically within the Colab session. Google Drive was mounted to provide persistent storage for the dataset, trained models, and output artefacts across sessions.

5.2 Dataset Acquisition

The DICOM-LIDC-IDRI-Nodules dataset was downloaded using the NBIA Data Retriever tool provided by The Cancer Imaging Archive. A `.tcia` manifest file was obtained from the TCIA portal, opened in the NBIA Data Retriever, and the approximately 2.51 GB dataset was downloaded to local storage before being uploaded to Google Drive for Colab access.

5.3 SR File Parsing: Tabular Feature Extraction

DICOM SR files store radiologist annotations as coded values within a hierarchical `ContentSequence` tree. A recursive tree-walking algorithm was implemented to traverse this structure and extract the nine radiomic feature values. Numeric concept codes are mapped to integer scores (1–5) using a custom decoding function. Features from all four radiologists are averaged per nodule, and binary labels are assigned based on the consensus malignancy score. The extraction process yielded 448 records saved to `tabular_features.csv`.

5.4 SEG File Processing: Image Patch Extraction

DICOM SEG files encode three-dimensional binary segmentation masks as multi-frame pixel arrays. The extraction pipeline reads the pixel array, identifies non-empty frames where the segmentation is present, selects the middle non-empty slice as a representative two-dimensional cross-section, crops the nodule bounding box with a 12-pixel padding margin, and resizes the resulting patch to a standardised 64×64 pixel resolution. In total,

3,219 patches were extracted and stored as NumPy arrays alongside their corresponding binary labels.

5.5 Tabular Pipeline

5.5.1 Preprocessing

The feature matrix $\mathbf{X}$ of shape 448×8 and label vector $\mathbf{y}$ of shape 448 are prepared for modelling. Missing values arising from incomplete radiologist annotations — specifically in the *internalStructure* (3 missing) and *calcification* (6 missing) columns — are imputed with column medians. All features are subsequently normalised using `StandardScaler` to achieve zero mean and unit variance, ensuring that no single feature dominates the learning process due to scale differences.

5.5.2 Imbalance Handling Strategies

The dataset exhibits a 2:1 class imbalance (299 benign vs. 149 malignant nodules). Three strategies were implemented and evaluated, as summarised in Table 5.1.

Table 5.1: Imbalance Handling Strategies Implemented

Strategy	Method	Description
No Resampling	None	Train on the original imbalanced data (149 malignant, 299 benign)
SMOTE	Synthetic Minority Oversampling	Generates synthetic malignant samples by interpolating between existing minority class samples in feature space
BorderlineSMOTE	Borderline variant of SMOTE	Focuses synthetic sample generation on borderline (hard-to-classify) minority instances near the decision boundary
CTGAN	Conditional Tabular GAN	Deep learning generative model trained on the minority class; generates 150 synthetic samples to balance to 299 each

5.5.3 Model Training

Three classifiers were trained and evaluated in combination with each imbalance handling strategy using five-fold stratified cross-validation:

- **Random Forest:** 100 decision trees; `random_state=42`.
- **Gradient Boosting Machine:** 100 estimators; `random_state=42`.
- **Logistic Regression:** `max_iter=1000`; `random_state=42`.

Within each cross-validation fold, the `StandardScaler` is fitted on the training partition and applied to the test partition to prevent data leakage. Class resampling is applied exclusively within the training partition.

5.5.4 Evaluation Metrics

Each fold produces four metrics: AUC-ROC, Recall (Sensitivity), F1-Score, and Precision. Final reported results represent the mean across all five folds for each strategy–model combination. Recall is accorded particular importance in the medical context, as false negatives — missed malignancies — carry greater clinical risk than false positives.

5.6 CTGAN Synthetic Data Generation

The Conditional Tabular GAN (CTGANSynthesizer) from the Synthetic Data Vault (SDV) library was trained on the 149 malignant nodule feature vectors for 300 epochs. Following training, 150 synthetic malignant samples were generated and combined with the original dataset, producing a balanced dataset of 598 records (299 malignant, 299 benign) saved to `tabular_ctgan_balanced.csv`.

5.7 SHAP Explainability

SHAP TreeExplainer was applied to a Random Forest model trained on the full training set. Two visualisation types were generated: a beeswarm plot displaying the distribution of SHAP values for each feature across all samples — with colour indicating feature magnitude — and a bar plot ranking features by their mean absolute SHAP value, representing global feature importance. Both plots were saved to the Google Drive output directory.

5.8 Image Pipeline

5.8.1 Dataset Split

The 3,219 nodule patches were partitioned into a training set (80%; 2,575 patches) and a test set (20%; 644 patches) using stratified splitting to preserve the class ratio across both partitions.

5.8.2 Data Augmentation

To increase sample diversity and reduce overfitting, the training set was augmented using random horizontal flips and random vertical flips, each applied with 50% probability per sample. No augmentation was applied to the test set.

5.8.3 ResNet18 Architecture and Training

A ResNet18 model pretrained on ImageNet was adapted for binary nodule classification by replacing the final fully connected layer with a two-output linear layer. All pretrained weights were retained and fine-tuned during training. The training configuration is as follows:

- **Optimiser:** Adam ($\text{lr} = 10^{-4}$, $\text{weight decay} = 10^{-5}$)
- **Loss Function:** Cross-Entropy Loss
- **Learning Rate Scheduler:** StepLR (step size = 7, $\gamma = 0.5$)
- **Epochs:** 25
- **Batch Size:** 64
- **Hardware:** NVIDIA T4 GPU (Google Colab)

The model checkpoint with the highest validation AUC was saved for subsequent evaluation and deployment.

5.8.4 Grad-CAM Visualisation

Gradient-weighted Class Activation Mapping was applied to the best saved model checkpoint to produce spatial attribution heatmaps. The target convolutional layer is the final layer of `model.layer4`. Gradients flowing back from the predicted class score with respect to this layer's activation maps are globally average-pooled to obtain channel weights. A weighted linear combination of the activation maps is then computed and passed through a ReLU activation to produce the final heatmap, which is overlaid on the original nodule patch for visual inspection.

Chapter 6

Results and Analysis

6.1 Tabular Pipeline Results

Table 6.1 presents the mean five-fold cross-validation performance across all nine strategy–model combinations. Results are sorted by imbalance handling strategy.

Table 6.1: Tabular Pipeline — 5-Fold Cross-Validation Results

Strategy	Model	AUC	Recall	F1	Precision
BorderlineSMOTE	Gradient Boosting	0.878	0.725	0.704	0.701
BorderlineSMOTE	Logistic Regression	0.839	0.859	0.701	0.593
BorderlineSMOTE	Random Forest	0.870	0.657	0.657	0.668
No Resampling	Gradient Boosting	0.880	0.664	0.695	0.745
No Resampling	Logistic Regression	0.856	0.664	0.667	0.679
No Resampling	Random Forest	0.869	0.624	0.659	0.709
SMOTE	Gradient Boosting	0.875	0.705	0.708	0.724
SMOTE	Logistic Regression	0.849	0.784	0.681	0.606
SMOTE	Random Forest	0.868	0.671	0.673	0.687

6.2 Key Findings: Tabular Pipeline

The tabular pipeline experiments reveal several clinically and statistically important patterns. The highest AUC-ROC of 0.880 was achieved by the No Resampling + Gradient Boosting combination, demonstrating that the underlying radiomic features carry strong discriminative signal even without class balancing. The highest Recall of 0.859 was achieved by BorderlineSMOTE + Logistic Regression, confirming that BorderlineSMOTE is the most effective strategy for minimising false negatives — the most critical error type in cancer screening. SMOTE + Gradient Boosting yielded the best F1-Score of 0.708, representing the most balanced trade-off between precision and recall.

Gradient Boosting consistently achieved the highest AUC across all three imbalance strategies, reflecting its effectiveness as an ensemble method for structured tabular data. Logistic

Regression, despite its relative simplicity, benefited most from oversampling strategies in terms of recall improvement, suggesting that the class boundary in feature space is approximately linear. CTGAN successfully generated 150 synthetic malignant feature vectors, expanding the minority class representation from 149 to 299 samples and achieving a balanced dataset for training.

6.3 Image Pipeline Results

Table 6.2 presents the performance of the ResNet18 model on the held-out test set.

Table 6.2: Image Pipeline — ResNet18 Test Set Performance

Metric	Value
Best AUC-ROC	0.7400 (at Epoch 10)
F1-Score	0.3945
Recall	0.2966
Precision	0.5263
Training Time	≈4 minutes (T4 GPU)
Best Epoch	10 of 25

6.4 Pipeline Comparison

Table 6.3 provides a direct comparison of the best-performing configurations of each pipeline across all four evaluation metrics.

Table 6.3: Dual-Pipeline Performance Comparison

Metric	Best Tabular	ResNet18 CNN	Winner
AUC-ROC	0.880	0.740	Tabular (+14%)
Recall	0.859	0.297	Tabular (+56%)
F1-Score	0.708	0.395	Tabular (+31%)
Training Time	≈10 min	≈4 min (GPU)	CNN
Explainability	SHAP (feature scores)	Grad-CAM (heatmap)	Both

6.5 Discussion: Performance Difference Between Pipelines

The tabular pipeline substantially outperforms the image pipeline across all predictive metrics. This performance differential is primarily attributable to the nature of the input data available to the image pipeline. The SEG files used in this project encode binary segmentation masks — that is, white-on-black shape outlines of the nodule — rather than the actual grayscale CT scan intensities. Consequently, the ResNet18 model is trained on shape information alone, without access to the rich textural and density information present in real CT images. The radiomic features extracted from SR files, by contrast, encode expert-assessed quantitative descriptors that directly capture malignancy-relevant characteristics such as spiculation, texture, and lobulation.

It is anticipated that if the full 133 GB CT scan dataset were used to extract actual grayscale nodule patches, the CNN pipeline’s performance would improve substantially — potentially surpassing the tabular pipeline — consistent with results reported in the broader deep learning radiology literature.

Chapter 7

Web Application

7.1 Overview

The trained models from both pipelines are deployed as a live, publicly accessible web application built using Streamlit and hosted on Streamlit Cloud. The application is permanently available at:

`https://lung--cancer--prediction.streamlit.app`

The application is designed to serve radiologists, researchers, and medical students by providing an intuitive interface for real-time malignancy prediction with interpretable outputs.

7.2 Application Structure

The application is organised into two tabs, each corresponding to one pipeline. Table 7.1 summarises the structure and functionality of each tab.

Table 7.1: Web Application Tab Structure

Tab	Name	Input	Output
1	Tabular Input — Feature Scores	8 radiomic feature sliders (1–5 scale)	Prediction, confidence, probability chart, feature importance chart
2	Image Input — Nodule Image	Upload JPG/PNG nodule image	Prediction, confidence, probability chart, Grad-CAM visualisation

7.3 Tab 1: Tabular Input

The first tab presents a set of eight interactive sliders, each corresponding to one radiomic feature. The user — typically a radiologist reviewing a CT scan — adjusts each slider to reflect the feature score observed for the nodule under examination. The feature ranges and clinical guidance are consistent with the LIDC-IDRI annotation schema described in

Chapter 3. Upon submission, the application outputs a binary malignancy verdict (Malignant or Benign) alongside a confidence percentage, a malignancy probability score, a horizontal bar chart depicting class probabilities, and a feature importance bar chart derived from the trained Random Forest model.

Table 7.2 summarises the slider ranges and their clinical interpretations.

Table 7.2: Tabular Input Sliders: Ranges and Clinical Guidance

Feature	Range	Clinical Guidance
Subtlety	1–5	1 = Extremely subtle; 5 = Obvious nodule
Internal Structure	1–4	1 = Soft tissue; 2 = Fluid; 3 = Fat; 4 = Air
Calcification	1–6	1 = Absent; 6 = Eccentric calcification
Sphericity	1–5	1 = Linear; 5 = Round
Margin	1–5	1 = Poorly defined; 5 = Sharp
Lobulation	1–5	1 = No lobulation; 5 = Marked lobulation
Spiculation	1–5	1 = No spiculation; 5 = Marked spiculation
Texture	1–5	1 = Non-solid (GGO); 5 = Solid

7.4 Tab 2: Image Input

The second tab accepts an uploaded JPG or PNG nodule image. The image may be a cropped CT scan patch or a segmentation mask patch exported from the dataset. The application internally resizes the image to 224×224 pixels, converts it to a three-channel tensor, and passes it through the loaded ResNet18 model. The outputs include a binary verdict, a confidence percentage, class probabilities, and a three-panel Grad-CAM visualisation showing the original image, the heatmap, and the overlay — enabling users to visually inspect which spatial regions of the nodule influenced the prediction.

7.5 File Structure and Model Loading

The application repository on GitHub contains the main Streamlit application script (`app.py`), the trained Random Forest model (`rf_model.pkl`), the fitted StandardScaler (`scaler.pkl`), and a dependency specification file (`requirements.txt`). The ResNet18 model weights

file (43 MB) is hosted on Google Drive and automatically downloaded to the deployment server on first launch using the `gdown` library, avoiding GitHub's file size constraints.

Chapter 8

Conclusion

This project successfully designed, implemented, and deployed a comprehensive dual-pipeline artificial intelligence system for lung cancer malignancy prediction using the LIDC-IDRI dataset. The following key findings and contributions are drawn from the experimental work.

The tabular machine learning pipeline, leveraging radiomic features extracted from DICOM SR files, achieved a peak AUC-ROC of 0.880 and a recall of 0.859 using BorderlineSMOTE with Logistic Regression, demonstrating strong predictive performance with clinically meaningful interpretability through SHAP analysis. Among the three imbalance handling strategies evaluated, BorderlineSMOTE was the most effective for improving recall — the most clinically consequential metric in cancer screening, where false negatives carry greater risk than false positives. CTGAN successfully augmented the minority class from 149 to 299 samples, providing a balanced training set for downstream experiments.

The image pipeline using ResNet18 achieved an AUC-ROC of 0.740, a result that is fundamentally constrained by the binary nature of the segmentation mask inputs rather than the model architecture or training procedure. This observation provides a clear and actionable direction for future improvement through the use of actual CT scan intensity patches.

The project contributes an end-to-end, reproducible, and publicly deployed system that advances the practical application of AI in radiological decision support. The live Streamlit application demonstrates that such systems can be made accessible to clinicians without requiring specialised technical infrastructure.

Chapter 9

Future Scope

Several promising directions exist for extending and improving the system developed in this project.

The most impactful near-term improvement would be to train the CNN pipeline on grayscale nodule patches extracted from the full 133 GB CT scan archive rather than binary segmentation masks. Access to CT intensity information would provide the model with texture, density, and morphological cues that are highly discriminative for malignancy, with expected AUC improvements to 0.85 or above based on comparable studies in the literature.

An ablation study comparing EfficientNet-B0 against ResNet18 would provide additional insight into the effect of model architecture on performance in this domain. Furthermore, a three-dimensional CNN — such as 3D-ResNet — could leverage the full volumetric information present across all CT slices of a nodule, rather than relying on a single representative two-dimensional slice.

The web application could be enhanced with an integrated DICOM viewer allowing radiologists to upload full CT scans, with automatic nodule detection and cropping as a preprocessing step. Longitudinal patient tracking — storing prediction histories with timestamps for monitoring nodule progression over time — would further increase clinical utility.

From a modelling perspective, ensemble fusion of the tabular and image pipeline predictions into a single combined output could improve overall accuracy and robustness. Monte Carlo Dropout could be incorporated to provide calibrated prediction uncertainty estimates, giving clinicians a measure of confidence in individual predictions. Training on additional datasets such as LUNA16 would improve generalisation across different scanner manufacturers and imaging protocols.

Finally, a mobile application targeting point-of-care use in resource-limited clinical settings would extend the system’s reach beyond hospital-based imaging infrastructure.

Chapter 10

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Live Application: <https://lung--cancer--prediction.streamlit.app>

Github Repository Link:

https://github.com/vaidaadya19-prog/Lung_Cancer_Prediction